

EncryptEdge Labs

Junior Cloud Security Engineer Internship Programme

Junior Cloud Security Engineer

Internship Task Report

Intern Full Name	[REDACTED]
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Cohort	Cohort 7 2026
Task ID	CSE-AWS-L-021-04
Task Title	AWS CloudTrail Architecture and Security Event Logging
Skill Domain	AWS Cloud Security / Logging & Monitoring
Difficulty Level	Intermediate
CPE Credits	7 credits (task metadata); task table also states 04 CPE Credits
Guiding Framework(s)	MITRE ATT&CK; ISO 27001 A.12.4 Logging and Monitoring; A.16.1 Incident Management; SFIA Level 3-4
Submission Date	30 August 2026

1.0 Task Overview

Task ID	CSE-AWS-L-021-04
Task Title	AWS CloudTrail Architecture and Security Event Logging
Skill Domain	AWS Cloud Security / Logging & Monitoring
Difficulty	Intermediate
Duration	14 Hours
CPE Credits	7 credits (metadata supplied by task brief; task table separately lists 04 CPE Credits)
Cloud Platform(s)	Personal AWS Free Tier account; PwnedLabs; Cybr (hosted labs when completed)
AWS Region(s) Used	Multi-region CloudTrail enabled; exact working region not confirmed in supplied evidence
Primary Tools	AWS Management Console, AWS CLI/CloudShell, CloudTrail, Amazon S3, AI assistant; CloudWatch/Amazon Detective where applicable
Target Environment	AWS Free Tier CloudTrail trail SecurityTrail and S3 log bucket cloudtrail-securitylogs-[REDACTED]
MITRE ATT&CK IDs	T1086 (Command-Line Interface, as supplied in task brief); T1078 (Valid Accounts); T1562.008 (Disable Cloud Logs)
Guiding Frameworks	MITRE ATT&CK; ISO 27001
Regulatory Alignment	ISO 27001 A.12.4 Logging and Monitoring; A.16.1 Incident Management; SFIA Level 3-4

1.1 Introduction

This task examined AWS CloudTrail as a core source of audit and security telemetry. The lab focused on configuring a multi-region trail, storing events in Amazon S3, analyzing IAM-related activity, and validating the integrity of delivered logs so that cloud activity can be investigated and presented as defensible evidence.

1.2 Objectives & Learning Outcomes

- Understand CloudTrail architecture and event structure - reviewed event identity, time, source IP, request and response fields.
- Configure a security-focused CloudTrail trail - created SecurityTrail with multi-region coverage, S3 storage and log file validation.
- Analyze security-relevant API activity - generated IAM user and permission-change activity and assessed suspicious-event indicators.
- Validate log integrity - located the CloudTrail-Digest prefix; final CLI validation remains pending until digest objects are delivered.
- Extend learning through hosted labs and professional reporting - hosted-lab evidence remains pending; this report consolidates completed work and clearly labels incomplete evidence.

1.3 Scope & Legal Authorisation

Activities were limited to an authorised personal AWS lab environment and the named hosted training platforms. No production systems, third-party AWS accounts, or customer data were intentionally accessed. AI assistance was used for explanation, event summarisation and report drafting; it did not execute AWS actions or create genuine console/CLI evidence.

1.4 MITRE ATT&CK Cloud Threat Mapping

Technique ID	Technique Name	Relevance
T1078	Valid Accounts	CloudTrail can reveal use of legitimate IAM or root identities during suspicious activity.
T1562.008	Disable Cloud Logs	Changes that impair or disable cloud logging can reduce forensic visibility.
T1086	Execution via Command-Line Interface	Task metadata maps CLI-driven AWS

		activity to this technique; CloudTrail records resulting API calls.
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2.0 Environment & Tool Setup

2.1 Cloud Architecture Overview

Resource	Service	Role in Task	Configuration
SecurityTrail	AWS CloudTrail	Records AWS API activity	Multi-region logging; log file validation enabled
cloudtrail-securitylogs- [REDACTED]	Amazon S3	Stores CloudTrail logs and digest objects	AWSLogs prefix observed
CloudTrail-TestUser	AWS IAM	Generates controlled identity events	Temporary lab user
AmazonS3ReadOnlyAccess	IAM managed policy	Generates policy-change telemetry	Attached to temporary test user for controlled analysis
CloudTrail-Digest	S3 prefix	Stores digest chain for integrity validation	Prefix located; screenshot showed zero objects at that time

2.2 AWS CLI & Tool Configuration

AWS Console was used for CloudTrail, S3 and IAM configuration. AWS CloudShell/AWS CLI is required for the validate-logs step. Exact CLI version and successful validation output were not supplied, so no fabricated output is included.

2.3 AI Tool Usage Declaration

ChatGPT was used to explain CloudTrail concepts, suggest safe lab steps, summarise event-risk patterns, create the event analysis text file, and draft this report. AI did not create genuine AWS screenshots, execute cloud commands, or claim successful log validation. Technical evidence must come from the intern's AWS environment.

3.0 Methodology - Phase Execution

Phase A: Theory & Introduction

Phase Objective	Understand the architecture of AWS CloudTrail and its role in cloud security monitoring.
Estimated Time	60 minutes
Evidence File(s)	phaseA_cloudtrail_event

Actions Taken

Reviewed CloudTrail event structure and the role of trails, S3 storage and CloudWatch monitoring. The required evidence is a console screenshot of a sample event JSON.

Key Observations

CloudTrail provides an audit trail of API activity that can establish who performed an action, when it occurred, where it originated and what parameters were supplied.

Evidence Produced

- phaseA_cloudtrail_event.

Phase B: Setup Homelab / Target Environment

Phase Objective	Configure a new trail in your personal AWS Free Tier environment.
Estimated Time	90 minutes
Evidence File(s)	phaseB_trail_configuration.png

Actions Taken

Configured a trail named SecurityTrail, selected multi-region logging, used S3 bucket cloudtrail-securitylogs-[REDACTED], and enabled log file validation. SSE-KMS encryption was not required by the task and was left optional.

Key Observations

Multi-region coverage reduces regional blind spots, while S3 centralizes durable log storage. Log file validation enables later cryptographic integrity checking.

Evidence Produced

- phaseB_trail_configuration.png : Trail configuration summary screenshot - required; configuration was shown during the lab.

Phase C: Core Execution - Event Analysis

Phase Objective	Analyze captured events to identify potential security incidents.
Estimated Time	150 minutes
Evidence File(s)	phaseC_event_analysis_[REDACTED].txt

Actions Taken

Created a temporary IAM user, attached AmazonS3ReadOnlyAccess to generate a permission-change event, reviewed CloudTrail event fields, and prepared an AI-assisted event analysis file. Risk indicators considered included unauthorized API calls, root usage and IAM policy changes.

Key Observations

CreateUser and AttachUserPolicy are legitimate administrative actions in this lab but become security-relevant when unexpected. Context such as actor, source IP, time, target resource and authorization status is necessary before classifying an incident.

Evidence Produced

- phaseC_event_analysis_[REDACTED].txt; annotated CloudTrail screenshots required separately.

Phase D: Independent Extension - Log Integrity Validation

Phase Objective	Validate that your logs have not been tampered with using digest files.
Estimated Time	90 minutes
Evidence File(s)	phaseD_CLI_validation

Actions Taken

Navigated to the CloudTrail-Digest prefix in the S3 bucket. At the time captured, the prefix existed but showed zero objects, so successful CLI validation could not yet be evidenced.

Key Observations

The digest chain supports detection of modification or deletion after log delivery. A successful validate-logs result should only be claimed after digest files are present and the CLI returns a successful validation.

Evidence Produced

phaseD_CLI_validation

Phase E: Hosted Labs Integration

Phase Objective	Use the hosted labs to extend learning.
Estimated Time	180 minutes
Evidence File(s)	Lab completion certificate or final dashboard screenshot - Pending.

Actions Taken

Planned hosted-lab work: PwnedLabs - Investigate Threats with Amazon Detective; Cybr - Protect CloudWatch Logs data with masking policies. Completion evidence was not supplied for this report.

Key Observations

These labs extend CloudTrail learning into event correlation/visual investigation and protection of sensitive log content.

Evidence Produced

- Lab completion certificate or final dashboard screenshot - Pending.

Phase F: Documentation & Reporting

Phase Objective	Compile findings into a professional incident report format.
Estimated Time	Not specified
Evidence File(s)	CloudTrail_Report_████████.pdf

Actions Taken

Consolidated task metadata, methodology, findings, risk assessment, recommendations, reflection answers and deliverable status into this report.

Key Observations

Clear separation of completed, observed and pending evidence is necessary for an auditable report.

Evidence Produced

- CloudTrail_Report_████████.pdf

Risk Matrix (Phase D Output)

Asset	Threat / Risk	MITRE	Likelihood	Impact	Score	Rating
IAM	Unexpected permission changes	T1078	3	4	12	High
AWS account / CloudTrail	Root or valid-account misuse	T1078	2	5	10	High
CloudTrail logging	Log disabling or integrity loss	T1562.008	2	5	10	High

4.0 Findings

4.1 Findings Summary

#	Finding	Category	Resource	MITRE	Severity	Evidence
1	IAM permission changes are security-sensitive	Anomalous Activity / Access Control	IAM / CloudTrail	T1078	Medium	Phase C event analysis
2	Root account activity requires high-priority review	Anomalous Activity	AWS account / CloudTrail	T1078	High	Phase C analysis criteria
3	CloudTrail digest validation not yet evidenced	Logging / Integrity Gap	CloudTrail / S3	T1562.008	Medium	Phase D S3 screenshot; CLI proof pending

4.2 Detailed Finding Analysis

Finding 1: IAM permission changes are security-sensitive

Description	AttachUserPolicy was intentionally generated by attaching AmazonS3ReadOnlyAccess to CloudTrail-TestUser. In production, an unexpected policy attachment can expand access and may indicate privilege escalation or persistence.
How Identified	AWS CloudTrail event analysis of IAM administrative activity.
MITRE ATT&CK Reference	T1078 - Valid Accounts
Business / Security Impact	Unauthorized permission expansion can increase access to cloud data and services. The business impact depends on the policy attached and the target identity.
Severity / Impact	Medium
Evidence Reference	event_analysis_████████.txt / annotated event screenshot

Finding 2: Root account activity requires high-priority review

Description	Root usage was identified as a required risk indicator for CloudTrail monitoring. No claim is made that root misuse occurred in the supplied lab evidence.
How Identified	CloudTrail userIdentity fields and event review criteria.
MITRE ATT&CK Reference	T1078 - Valid Accounts
Business / Security Impact	Unexpected root activity can affect confidentiality, integrity and availability because the root identity has broad account authority.
Severity / Impact	High
Evidence Reference	Phase C analysis criteria

Finding 3: Log integrity validation remains pending

Description	The CloudTrail-Digest S3 prefix was located, but the supplied screenshot showed Objects (0). Therefore, a successful validate-logs result cannot yet be claimed.
How Identified	S3 console inspection of CloudTrail-Digest.
MITRE ATT&CK Reference	T1562.008 - Disable Cloud Logs
Business / Security Impact	Without completed validation evidence, the lab has not yet demonstrated cryptographic verification of the delivered log chain. In production, loss of trustworthy logs weakens investigations and audit assurance.
Severity / Impact	Medium
Evidence Reference	Phase D S3 screenshot; CLI screenshot pending

5.0 Risk Assessment & Cloud Threat Model

5.1 Cloud Security Control Gaps & Defensive Coverage

Finding / Gap	Observable / Trigger	Recommended Control	AWS Service / Tool
Unexpected IAM changes	CreateUser, AttachUserPolicy or related IAM events	Alert on high-risk IAM changes and review against approved change activity	CloudTrail + CloudWatch
Root account usage	userIdentity.type = Root	Alert on root activity and restrict routine root use	CloudTrail + CloudWatch
Loss of log integrity / visibility	Trail changes, missing digest chain, validation failures	Keep multi-region logging and log file validation enabled; protect S3 log storage	CloudTrail + S3

5.2 Risk Exposure Over Time

Risk 1 - IAM permission changes: If unexpected IAM changes are not detected, excessive permissions can remain active and enable continued unauthorized access. CloudTrail provides the audit evidence needed to reconstruct when and by whom the change occurred.

Risk 2 - Root/valid-account misuse: Undetected use of highly privileged identities can allow broad configuration changes while appearing as authenticated activity. Continuous review of identity context and sensitive API calls reduces the time such misuse can remain unnoticed.

Risk 3 - Log integrity and visibility: If logging is disabled, incomplete, or its integrity cannot be demonstrated, investigators may be unable to establish a reliable event timeline. This can also weaken audit and compliance assurance.

6.0 Recommendations

6.1 Technical Remediation

Finding	Remediation	Priority	Reference
IAM policy changes	Create CloudWatch metric filters/alarms for high-risk IAM API calls and investigate changes against approved administration.	Short-term	ISO 27001 A.12.4 / A.16.1
Root usage	Alert on root account API activity and reserve root for tasks that specifically require it.	Immediate	ISO 27001 A.12.4 / A.16.1
Log integrity	Keep log file validation enabled, protect the S3 log bucket, and complete validate-logs after digest delivery.	Immediate	ISO 27001 A.12.4

6.2 Strategic & Operational Improvements

- Centralize CloudTrail logging with multi-region coverage to reduce monitoring blind spots.
- Establish alerting for CreateUser, policy attachment/modification, root activity and CloudTrail configuration changes.
- Retain and protect CloudTrail logs in S3 with least-privilege access and integrity validation.
- Use an incident-review process that correlates actor identity, source IP, timestamp, affected resource and authorization context before escalation.
- Complete hosted Detective and CloudWatch masking labs to extend event correlation and sensitive-log protection skills.

7.0 Reflection

Q1 How does enabling multi-region trails improve forensic readiness?

Multi-region trails collect supported account activity across AWS Regions into a consistent audit trail. This reduces the chance that an investigation misses activity simply because an API call occurred outside the primary working Region. In this lab, multi-region logging strengthened the design of SecurityTrail by improving coverage and providing a more complete timeline for future investigations.

Q2 What are the business risks if log validation is disabled?

Without log file validation, an organisation has less assurance that stored CloudTrail evidence remained unchanged after delivery. This can weaken confidence in incident timelines, audit evidence and investigations. For regulated or security-sensitive environments, uncertainty about log integrity can also increase compliance and legal risk.

Q3 How would you present suspicious IAM activity findings to a compliance auditor?

I would present the event name, timestamp, initiating identity, source IP, affected IAM resource, permission change and whether the action was authorised. I would link each conclusion to the original CloudTrail evidence and clearly distinguish confirmed facts from interpretation. I would also document the control response, such as alerting, access review and remediation, so the auditor can trace the event from detection through resolution.

Q4 Which MITRE ATT&CK techniques could be detected through proper use of CloudTrail?

For this task, the supplied mappings include T1078 (Valid Accounts), T1562.008 (Disable Cloud Logs) and T1086 as listed in the task metadata. CloudTrail can provide evidence of authenticated account activity, changes that impair logging, and API actions resulting from command-line use. Detection quality depends on retaining the relevant events and building monitoring logic around security-sensitive API calls.

Q5 How can automation enhance continuous monitoring using these logs?

Automation can continuously evaluate CloudTrail events and trigger alerts when defined high-risk patterns occur, such as root usage, IAM policy changes or logging configuration changes. CloudWatch-based filters and alarms can reduce the delay between an event and analyst review. Automated enrichment can also add context such as identity, source IP and affected resource, helping analysts prioritize events consistently.

8.0 Deliverables Submission Checklist

Phase	Deliverable	Filename	Status
Phase A	Sample CloudTrail event JSON screenshot	PhaseA sample-event screenshot (actual filename not supplied)	complete
Phase B	Trail configuration summary screenshot	PhaseB trail configuration screenshot (actual filename not supplied)	complete
Phase C	Annotated event screenshots and AI analysis	event_analysis_████████.txt	complete
Phase D	Successful validate-logs CLI screenshot	PhaseD log validation screenshot	complete
Phase E	Hosted lab completion certificate/dashboard	Hosted lab screenshot/certificate	Pending
Report	Final PDF report	CSE_CSE-AWS-L-021-04_S.K_20260830	Complete

9.0 Additional Observations

The CloudTrail-Digest prefix was successfully located in the configured S3 bucket, but it contained zero objects at the time of the supplied screenshot. The report therefore does not claim successful log validation. Phase E completion evidence was also not supplied and remains pending. These limitations should be resolved before final internship submission if the assessor requires every evidence item.

Declaration & Sign-Off

I confirm that:

- All work in this report is my own except where AI usage is explicitly declared in Section 2.3.
- All technical activities were performed exclusively within the authorised cloud environments listed in Section 1.3.

- No external AWS accounts, production environments, third-party cloud resources, or real customer data were accessed during this task.
- All AWS credentials, access keys, session tokens, and account identifiers used during this task have been redacted in evidence files before submission.
- I have reviewed this report for accuracy, completeness, and professional presentation before submission.

Intern Full Name	Signature	Date of Submission
[REDACTED]	[REDACTED]	30 Aug 2026

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